

Statistical Analysis of Audience Reviews for 3 Idiots

Using IMDb Audience Reviews Dataset

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- Introduction
- Methodology
- Statistical Analysis
- Visual Analysis
- Conclusion

The main objectives of this project are:

1. To analyze audience reviews of **3 Idiots**.
2. To measure audience satisfaction using statistical techniques.
3. To identify rating trends and review patterns.
4. To perform descriptive statistical analysis.
5. To visualize the collected data using charts and graphs.
6. To interpret audience behavior and movie popularity.

Dataset Information

Attribute	Description
Rows	2226
Columns	4
Rating Scale	1–10
Movie	3 Idiots

Dataset Structure

Column	Description
Username	Reviewer ID
Rating	Score (1–10)
Review Title	Review Heading
Review Text	Complete Review

Sample Dataset

Username	Rating	Review Title
Unknown	9	Wow this American loves this movie
Unknown	9	Hat-Trick Heroes
Unknown	9	Great Movie

Data Collection

- IMDb
- Rotten Tomatoes
- Letterboxd
- Google Reviews

Data Cleaning

- Remove duplicate reviews
- Handle missing values
- Convert ratings into numeric format
- Keep ratings between 1–10

Sentiment Classification

Rating Categories

Rating Range	Sentiment
8–10	Positive
5–7	Neutral
1–4	Negative

Purpose

- Simplify review analysis.
- Understand audience perception.
- Create visualizations and sentiment summaries.

Mean

$$\bar{x} = \frac{\sum x}{n}$$

Average audience rating.

Median

Middle rating after sorting.

Mode

Most frequently occurring rating.

Range

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Variance

$$\sigma^2 = \frac{\sum (x - \bar{x})^2}{N}$$

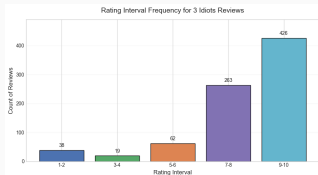
Standard Deviation

$$\sigma = \sqrt{\sigma^2}$$

Measures consistency of audience opinions.

Frequency Distribution

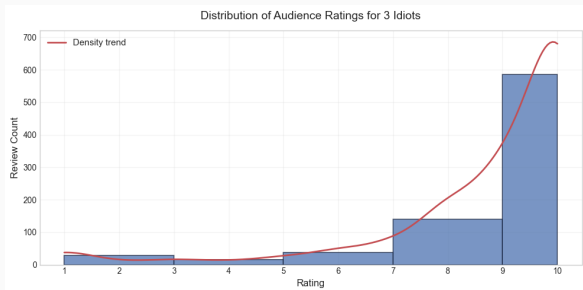
Rating Interval Frequency Distribution



Discussion:

- Highest frequency occurs in the **9–10** rating interval.
- Most viewers gave very high ratings.
- Ratings are concentrated in the upper intervals.
- This indicates strong audience appreciation for the movie.

Rating Distribution



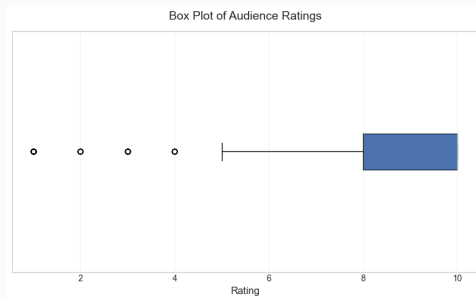
Discussion

- Ratings are concentrated near **9** and **10**.
- The distribution is **left-skewed**.
- Most viewers provided high ratings.
- Indicates strong positive audience reception.

Interpretation:

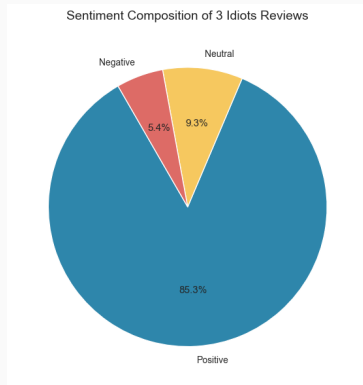
The histogram shows that audience ratings are heavily clustered at the higher end of the scale, suggesting that *3 Idiots* was highly appreciated by ¹⁰

Box Plot Analysis



Interpretation

- The median rating is very high, indicating strong audience satisfaction.
- Most ratings are concentrated in the upper rating range.
- A few low-rating outliers exist, representing viewers who disliked the movie.
- The majority of reviews are positive, showing widespread appreciation.



Key Observations

- Positive Reviews: 85.3%
- Neutral Reviews: 9.3%
- Negative Reviews: 5.4%

Interpretation

- The majority of audience reviews are positive.
- Only a small percentage of viewers gave negative reviews.
- Audience sentiment is overwhelmingly favorable.
- The movie is widely appreciated by viewers.